

# AETHYR VILLELA

972-374-7630 | [ajv12316@gmail.com](mailto:ajv12316@gmail.com) | [www.linkedin.com/in/ajv12316](http://www.linkedin.com/in/ajv12316)

## EDUCATION

**The University of Texas at Austin**, Austin, TX

May 2026

*Bachelor of Science, Biomedical Engineering, Concentration in Instrumentation & Imaging*

Certificate in Forensic Science

*Relevant Coursework:* Biomedical Engineering Signals and Systems Analysis, Micro and Nanotechnology for Sensors, Mechatronics

## SKILLS

**Technical:** Python, MATLAB, C++, R, Fusion360, SolidWorks, MS Office

**Certifications:** 3D Printing, Laser Cutting, IBM Project Manager

**Honors:** Friends of Alec Engineering Scholarship, FHL Bank of Dallas Scholarship Recipient, 2022 Hispanic Scholarship Foundation Scholar, ULN Scholar, Ramshorn Program Scholar, Texas Challenge Scholarship

## EXPERIENCE

**SurfPlasma**, *Business Development Intern*

Jun 2025 - Present

- Collaborate with R&D and commercialization leadership to align the product roadmap with market needs and regulatory pathways
- Support regulatory strategy by organizing documentation in compliance with ISO, FDA, OSHA, and EPA standards
- Conduct strategic outreach to identify and engage potential B2B partners, generating leads for early commercialization

**Project Advance Austin (PAA)**, *Project Management Intern*

Aug 2024 - May 2025

- Collaborated with SpineHope and a team of 3 other UT students to design an educational curriculum, website, and interactive activities to simplify spinal conditions for children and promote body positivity
- Earned IBM Project Manager Certification, showcasing expertise in managing project timelines, tasks, and communication effectively

**Crown Castle**, *AI/ML May Intern*

May 2024

- Developed a computer vision AI/ML model using Python, Docker, Linux, and TensorFlow to detect brain tumors with a 71% validation rate
- Utilized a dataset of annotated MRI images and implemented a convolutional neural network (CNN) architecture

**Worcester Polytechnic Institute (WPI)**, *Engineering Design Intern*

May 2023 - Jul 2023

- Collaborated with a team of 3 biomedical engineering students to design solutions for global health challenges
- Conducted market research, finding a needs gap and investigating
- Developed a Portable Water Purification System by utilizing SolidWorks

## PROJECTS

**2025 National Medical Device Make-A-Thon Competition**

Feb 2025

- Placed 6th nationwide and won First Place prelim round with a low-intensity ultrasound device for osteoarthritis; delivered regulatory and market analysis

## LEADERSHIP & COMMUNITY INVOLVEMENT

**Biomedical Engineering Society (BMES)**, *Vice President External*

May 2024 - Present

- Spearheaded a new program, BMESim, enabling BMES members to view, participate, and conduct outreach through Dell Children's medical simulation program
- Executed a Case Competition – increased team participation by 67% compared to last year
- Organized a Career Night panel, contributing to the professional development of over 150 BMES members

**Ramshorn Scholars Program (RSP)**, *Manager*

May 2024 - Present

- Oversee a team of 45 students, providing guidance to 150 first-year engineers while applying project management and process improvements to enhance program outcomes
- Assist voter registration of Ramshorn Scholars through directed initiatives and collaborations
- Organize 6 socials throughout the academic year, increasing engagement and student retention